

1. Implement a program (shell script) that allows a user to manage student records by taking Student Roll Number and Student Name as input. The program should perform basic CRUD operations (Create, Read, Update, Delete) and store all records in a file for persistent storage.
2. Implement the C program in which the main function accepts the integers to be sorted. Main function uses the FORK system call to create a new process called a child process. Parent process sorts the integers using a sorting algorithm and waits for the child process using WAIT system call to sort the integers using any sorting algorithm. Also demonstrate zombie and orphan states. . Use basic programming constructs only; do not use advanced inbuilt library functions
3. Implement the C/C++ program to simulate CPU Scheduling Algorithm round robin (preemptive) with different arrival times. Use basic programming constructs only; do not use advanced inbuilt library functions
4. Implement the C/C++ program to simulate CPU Scheduling Algorithm Priority (preemptive) with different arrival times. . Use basic programming constructs only; do not use advanced inbuilt library functions
5. Implement the C/C++ program to simulate CPU Scheduling Algorithm Shortest Job Remaining First (preemptive) with different arrival times. Use basic programming constructs only; do not use advanced inbuilt library functions
6. Implement the C/C++ program to simulate CPU Scheduling Algorithm FCFS (non-preemptive) with different arrival times. . Use basic programming constructs only; do not use advanced inbuilt library functions
7. Implement the C/C++ program to simulate CPU Scheduling Algorithm shortest job first (non-preemptive) with different arrival times. Use basic programming constructs only; do not use advanced inbuilt library functions
8. Implement the C/C++ program to simulate CPU Scheduling Algorithm priority (non-preemptive) with different arrival times. Use basic programming constructs only; do not use advanced inbuilt library functions
9. Implement the C program for bounded buffer producer- consumer problem.
10. Implement the C/C++ program for LRU Page Replacement Algorithm with n frames as an input. Use basic programming constructs only; do not use advanced inbuilt library functions
11. Implement the C/C++ program for Optimal Page Replacement Algorithm with n frames as an input. Use basic programming constructs only; do not use advanced inbuilt library functions

12. Implement the C/C++ program for FIFO Page Replacement Algorithm with n frames as an input. Use basic programming constructs only; do not use advanced inbuilt library functions
13. Implement the C/C++ program for FCFS disk scheduling with a given request queue (at least 15 requests) and initial disk head position. Use basic programming constructs only; do not use advanced inbuilt library functions
14. Implement the C/C++ program for SSTF disk scheduling with a given request queue (at least 15 requests) and initial disk head position. Use basic programming constructs only; do not use advanced inbuilt library functions
15. Implement the C/C++ program for LOOK disk scheduling with a given request queue (at least 15 requests) and initial disk head position. Use basic programming constructs only; do not use advanced inbuilt library functions
16. Implement the C/C++ program for CLOOK disk scheduling with a given request queue (at least 15 requests) and initial disk head position. Use basic programming constructs only; do not use advanced inbuilt library functions
17. Implement the C/C++ program for SCAN disk scheduling with a given request queue (at least 15 requests) and initial disk head position. Use basic programming constructs only; do not use advanced inbuilt library functions
18. Implement the C/C++ program for CSCAN disk scheduling with a given request queue (at least 15 requests) and initial disk head position. Use basic programming constructs only; do not use advanced inbuilt library functions
19. Implement the C/C++ program in which the main function accepts the integers to be sorted. Main function uses the FORK system call to create a new process called a child process. Parent process sorts the integers using a sorting algorithm and waits for the child process using WAIT system call to sort the integers using any sorting algorithm. Also demonstrate zombie and orphan states. . Use basic programming constructs only; do not use advanced inbuilt library functions
20. Implement the C program for bounded buffer producer- consumer problem.
21. Implement the C/C++ program for LRU Page Replacement Algorithm with n frames as an input. Use basic programming constructs only; do not use advanced inbuilt library functions

22. Implement the C program in which the main function accepts the integers to be sorted. Main function uses the FORK system call to create a new process called a child process. Parent process sorts the integers using a sorting algorithm and waits for the child process using WAIT system call to sort the integers using any sorting algorithm. Also demonstrate zombie and orphan states. Use basic programming constructs only; do not use advanced inbuilt library functions
23. Implement the C program for bounded buffer producer- consumer problem.
24. Implement a program (shell script) that allows a user to manage student records by taking Student Roll Number and Student Name as input. The program should perform basic CRUD operations (Create, Read, Update, Delete) and store all records in a file for persistent storage.
25. Implement a program (shell script) that allows a user to manage train records by taking train Number and train Name as input. The program should perform basic CRUD operations (Create, Read, Update, Delete) and store all records in a file for persistent storage.